DVLSI-2025 Lab Assignment-2

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***Abstract*—This assignment report presents a simulation-based study of NMOS and PMOS transistors with 45nm technology parameters. Its key analyses include the Ids variations with voltage, device geometry, and temperature, highlighting non ideal effects, threshold voltage shifts, and scaling trends which are relevant for advanced CMOS design. By this we get insights into transistor performance differences and behavior under varied operating conditions.**

***Index Terms*—NMOS transistor, PMOS transistor, threshold voltage, sub-threshold slope.**

INTRODUCTION

MOSFETs are fundamental building blocks of digital systems and model IC design. As we scale down its dimensions it becomes significant to understand their behaviour for relible designing. This report focuses on analysing MOSFETs built in 45 nm technology to study the changes in drain current, voltage and device geometry. Threshold voltage, sub-threshold slope and scaling of aspect ratios are key parameters that provide insights into transistor performance.

I. ANSWER 1

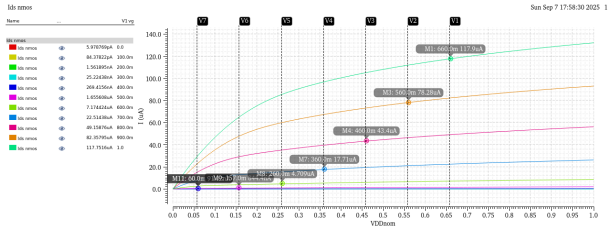


Fig. 1. Ids vs Vds (varying Vgs from 0 to Vdd)

As Vgs increases it can be observed that Ids increases significantly indicating stronger band formation with increase in charge carrier mobility for higher Vgs.The saturation of drain current is expected after Vds crosses the overdrive voltage. However, a slopy curve is seen post saturation. This occurs as a result of channel length modulation. 1 The current

I_{DS}-NMOS	I_{DS}-PMOS	Ratio
215.34523μA	152.1135μA	1.4156

ration of nmos to pmos is larger than 1. Hence, we can observe that nmos is stronger than pmos by a factor of 1.4156.

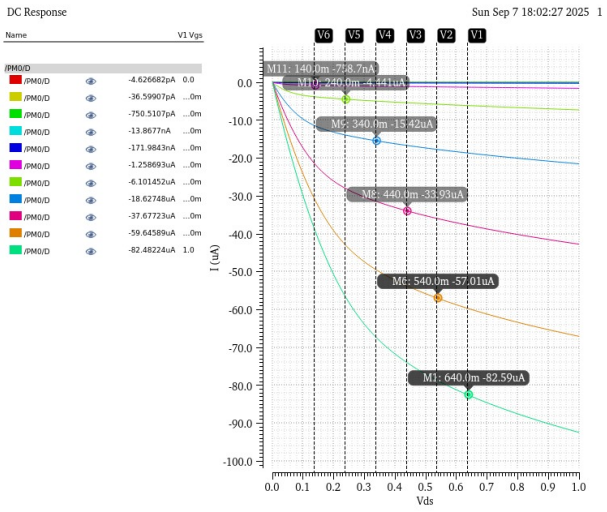


Fig. 2. Ids vs Vds (varying Vgs from 0 to Vdd)

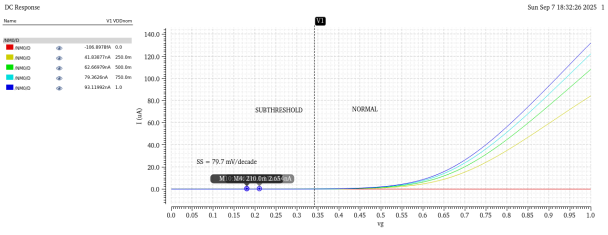


Fig. 3. Ids vs Vsg for varying Vsd (0 to 1 V)

II. ANSWER 2

The above figure 3 shows a slow increase in current when Vgs is below Vth. Whereas, as Vgs crosses Vth a rapid increase in current is visible. The subthreshold region is defined for Vgs ¡ Vth and the normal region is defined for Vgs ¿ Vth. The subthreshold slop is calculated and comes out to be approximately 79.7 mV/decade. It is calculated using the formula: $S = \frac{d(\log I_D)}{dV_{GS}}$
where Id is the drain current and Vgs is the gate-source voltage.

The plot shows a clear subthreshold region upto Vsg = 0.363 V which is the threshold voltage. For higher values of Vsg we get to see the normal region. The subthresold slope is approximated around 81.46 mV/decade. 4

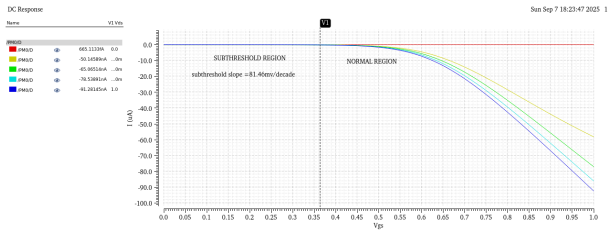


Fig. 4. Ids vs Vsg for varying Vsd (0 to 1 V)

III. ANSWER 3

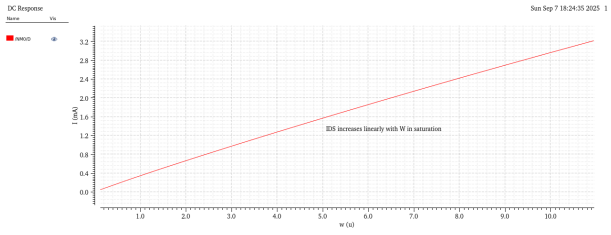


Fig. 5. Ids vs Vgs for varying Vds (0 to 1 V)

As per the shockley equation: $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T)^2$ Id varies linearly with W and a linear graph is expected as seen in the figure 5. The linear graph confirms the linear relation.

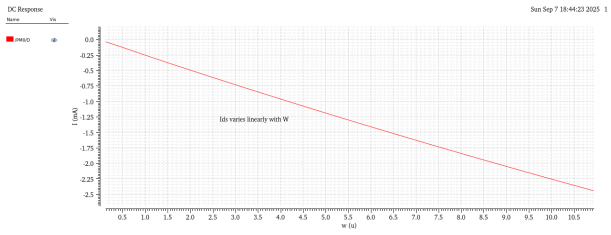


Fig. 6. Ids vs Vsg for varying Vsd (0 to 1 V)

Id varies linearly with W and a linear graph is expected as seen in the figure 6. The linear graph confirms the linear relation.

IV. ANSWER 4

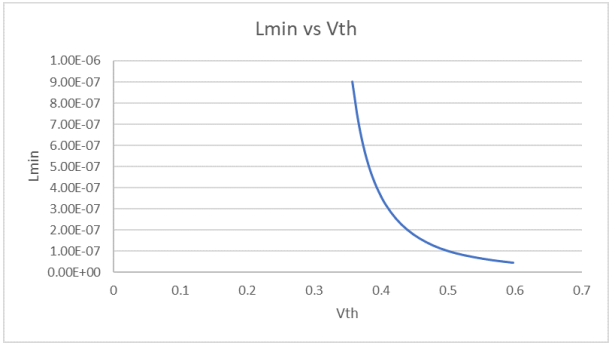


Fig. 7. Lmin vs Vth plot for NMOS

The plot 7shows that as the Vth increases, Lmin decreases sharply. This trend reflects short channel effects including drain induced barrier lowering (DIBL).

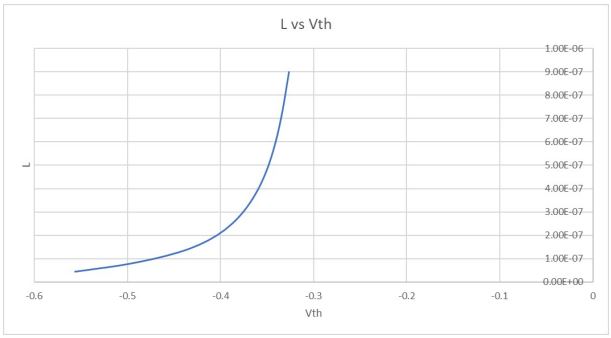


Fig. 8. Lmin vs Vth plot for PMOS

The PMOS plot (L vs Vth) shows that as the channel length increases, the threshold voltage becomes less negative and approaches zero.8

Increasing the channel length suppresses short-channel effects, resulting in more stable threshold voltage and reduced leakage current for PMOS devices.

V. ANSWER 5

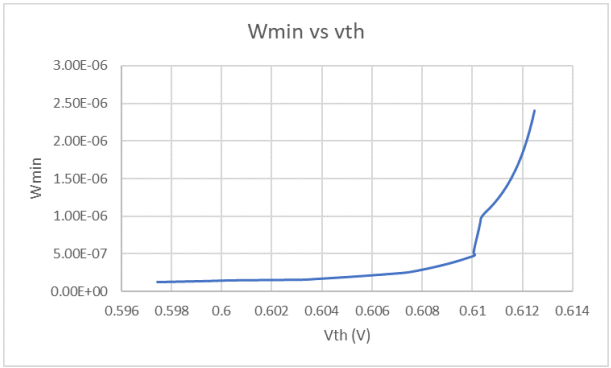


Fig. 9. Wmin vs. Vth for NMOS

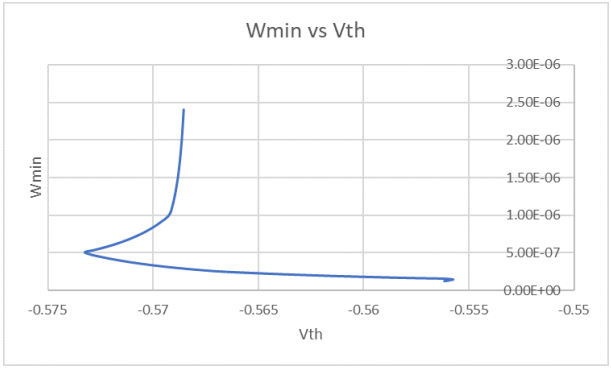


Fig. 10. Wmin vs. Vth for PMOS

The above plot 9shows that as Wmin increases, Vth rises slightly until a critical width is reached, where Vth rises

sharply. So vth is insensitive to small width variation but it is highly sensitive near the geometric limit.

Similar can be observed for the 10 pmos. The non linear behaviour suggests that scaling of width can impact threshold voltage.

VI. ANSWER 6

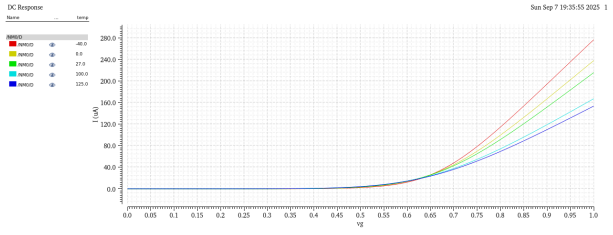


Fig. 11. Ids vs Vgs varying with temperature (NMOS)

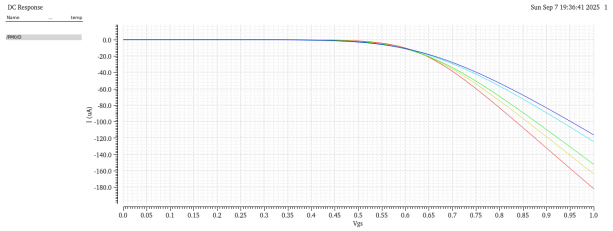


Fig. 12. Ids vs Vsg varying with temperature (PMOS)

The drain current decreases with increase in temperature due to reduced carrier mobility. The threshold voltage also decreases with temperature which results in increased sub threshold leakage. At low temperatures mobility is high and high Vth results in better transistor performance.11

Similarly, PMOS transistors show decrease in magnitude of Id with increase in temperature as can be observed in 12. This is due to high hole mobility degradation. Low temperature operation hence improves drive current and reduces leakage.

VII. Conclusion

Here, we simulated and analysed 45nm technology NMOS and PMOS transistors using Cadence Virtuoso Layout Suite tool. All the simulation results closely matched with that in the theory. Due to DRC violation Id vs W simulation showed some abnormal behaviour after 10µm. Abnormal behaviour is also shown by last two simulations.

References

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